

MECHWISE

Workshop Management System

Technical Specification Document

Version 1.0 — August 2026

Prepared for: Developer handover

First client: Dhalla Automotive Pty Ltd

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1. Executive summary

MechWise is a cloud-based SaaS platform for Australian mechanical workshops, designed to replace outdated PHP systems, spreadsheets, and manual processes with an integrated, modern workshop management solution. The platform covers the full operational lifecycle: client and vehicle management, service catalogues with structured pricing, quote-to-invoice workflow, parts inventory with reorder automation, appointment scheduling with bay management, SMS-driven service reminders, BAS/GST-compliant accounting, and a customer-facing portal.

The initial reference implementation and first paying customer is Dhalla Automotive Pty Ltd, an established workshop in Kingswood, Western Sydney. The platform is being designed to be productised for annual subscription sales to other workshops across Australia.

1.1 Business model

- **Starter (\$99/month):** Solo operators. Core CRUD, quotes, invoices, inventory, reminders. 1 user, 100 SMS/month.
- **Professional (\$199/month):** Growing workshops. Adds customer portal, floor board, dashboard analytics, BAS export. Up to 5 users, 500 SMS/month.
- **Enterprise (\$349/month):** Multi-location. Adds multi-workshop support, Xero/MYOB integration, consolidated reporting, white-label. Unlimited users, 2,000 SMS/month.

1.2 Current state of deliverables

This document consolidates the complete design work produced during the specification phase:

- **Database design:** 26-sheet enhanced Excel workbook populated with real Dhalla data (300 clients, 323 vehicles, 123 parts, 276 reminders, 11 staff).
- **SQL schema:** 20 relational tables with FK constraints, enums, CHECK constraints, 30+ indexes, and 6 pre-built dashboard views.
- **ERD diagrams:** 4 entity-relationship diagrams covering core entities, service catalogue, workflow pipeline, and operations/accounting.
- **Module specifications:** 8 deep-dive modules with field-level tables, business rules, validation rules, edge cases, and worked examples.
- **UI prototypes:** 12 interactive React/JSX screen prototypes demonstrating every major screen in the platform.
- **Production roadmap:** Architecture, tech stack, 4-phase development plan, Gantt timeline, 3 cost options, and pre-launch checklist.

2. System architecture

2.1 Architecture overview

Three-layer architecture with logical multi-tenant isolation via PostgreSQL Row-Level Security. All hosting in AWS Sydney (ap-southeast-2) for Australian data sovereignty.

Layer	Technology	Responsibility
Frontend	Next.js 14 + React 18 + Tailwind CSS	All 12 UI screens. SSR for customer portal. CSR for dashboard and workflow screens. Deployed to Vercel.
API	Node.js + Fastify 4 + Prisma 5	REST API with JWT auth. Workshop-scoped middleware. PDF generation. Rate limiting. Deployed to Railway or AWS ECS.
Database	PostgreSQL 15 on Supabase or RDS	20 tables, 30+ indexes, 6 views. RLS policies enforce tenant isolation. JSONB for flexible fields.
Storage	AWS S3 / Cloudflare R2	Invoice PDFs, logos, uploads. Workshop-prefixed paths. CDN-backed delivery.
SMS	MessageMedia or Twilio	Outbound reminders, confirmations, quote delivery. Inbound keyword parsing (YES, BOOK, STOP).
Email	Postmark or SendGrid	Transactional: invoices, quotes, password resets. SPF + DKIM for deliverability.
Payments	Stripe	Subscription billing. Plan management. Webhook integration.
Background	BullMQ + Redis	Scheduled tasks: reminder sends, quote expiry, overdue flagging, BAS rollover.
Monitoring	Sentry + Better Uptime	Error tracking, performance monitoring, uptime alerting.

2.2 Key architectural decisions

- **Monolith first:** Single Node.js API handles everything at launch. Microservices only after 50+ workshops with clear bottlenecks.
- **REST not GraphQL:** Well-defined screens with known data shapes. REST endpoints mapped 1:1 to screens are simpler to build, debug, and cache.
- **SSR for portal only:** Workshop dashboard is a private SPA. Customer portal benefits from SSR for load speed and link preview metadata.
- **Prisma with raw SQL escape:** Prisma handles 90% of queries. Complex reports and bulk operations use Prisma.\$queryRaw.

2.3 Multi-tenancy model

Shared database with logical isolation. Every table with business data includes a `workshop_id` column. PostgreSQL Row-Level Security (RLS) ensures queries are always filtered by the authenticated user's workshop. API middleware injects `workshop_id` from the JWT token into all queries automatically.

Workshop-scoped tables: `clients`, `vehicles`, `client_vehicles`, `staff`, `users`, `quotations`, `quote_lines`, `job_cards`, `job_card_lines`, `invoices`, `invoice_lines`, `payments`, `parts`, `purchase_orders`, `po_lines`, `appointments`, `service_reminders`, `maintenance_history`, `accounting_ledger`, `audit_log`.

Global tables (shared): `vehicle_makes`, `default job_categories`, `default part_categories`, `subscription_plans`.

3. Technology stack

Category	Choice	Rationale
Language	TypeScript (full stack)	One language across frontend and backend. Type safety. Shared types between API and components.
Frontend	Next.js 14 (App Router)	Server components for portal. File-based routing. Built-in API routes for webhooks.
CSS	Tailwind CSS 3	Utility-first. Consistent spacing and color tokens. Matches prototype design system.
Components	shadcn/ui + Radix	Accessible primitives. Copy-paste components with full control over styling.
State	Zustand + React Query	Zustand for UI state. React Query for server state with auto-caching and optimistic updates.
Backend	Fastify 4	2-3x faster than Express. Built-in schema validation. TypeScript-first.
ORM	Prisma 5	Type-safe queries. Migration system. Prisma Studio for visual data browsing.
Database	PostgreSQL 15	RLS for multi-tenancy. JSONB for flexible fields. Full-text search.
Caching	Redis (Upstash)	Session storage. BullMQ job queue. Report caching. Rate limiting.
PDF	React-PDF (@react-pdf/renderer)	Invoice PDFs as React components. Server-side generation.
Charts	Recharts	React-native charts for dashboard visualisations.
Testing	Vitest + Playwright	Vitest for unit/integration. Playwright for E2E testing.
CI/CD	GitHub Actions	PR checks (lint, type-check, test). Auto-deploy on merge.
Hosting	Vercel + Railway	Vercel for Next.js. Railway for API, Postgres, Redis. Both have Sydney regions. ~\$40-80/month.

4. Database schema

The complete PostgreSQL schema consists of 20 relational tables organised into four domains. The full SQL file (MechWise_Database_Schema.sql) is provided separately and should be used as the authoritative schema reference. Below is a summary of each table and its role.

4.1 Core entities

workshops

The root tenant table. Every record in the system belongs to a workshop.

Field	Type	Tags	Validation and notes
workshop_id	VARCHAR(10)	PK, Auto	Format: WS-001. Root key for multi-tenancy.
business_name	VARCHAR(200)	Required	Legal business name shown on invoices.
abn	VARCHAR(20)	Required, Validated	Australian Business Number. 11 digits. Required for ATO-compliant invoicing.
mvr_l_number	VARCHAR(20)		Motor Vehicle Repairer Licence. Required for e-safety inspections.
arc_number	VARCHAR(20)		Air Conditioning Refrigerant Certificate. Required for AC work.
address/phone/email	Various	Required	Workshop contact details. Appear on all customer-facing documents.
operating_hours	JSONB		Per-day hours object. Used by appointment scheduler and SMS send windows.
default_labour_rate	DECIMAL(8,2)		Default \$/hr for all job sub-items. Typically \$80-\$120 in Western Sydney.

clients

Customer records. Support both Individual and Business types. Progressive data capture — the system functions with minimal data and improves over time.

Field	Type	Tags	Validation and notes
client_id	VARCHAR(10)	PK, Auto	Format: CL-001. Sequential per workshop.
client_type	ENUM	Required	Individual or Business. Determines which name fields are shown.
first_name/last_name	VARCHAR(100)	Required	For individuals. Stored separately for sorting and search.
business_name	VARCHAR(200)		For business clients (fleet companies, government).
mobile_phone	VARCHAR(20)	Validated	Primary contact for SMS. Auto-formatted to 04XX XXX XXX.
email	VARCHAR(150)	Validated	Currently 0% captured in Dhalla data. Critical for portal access and email invoicing.
preferred_contact	ENUM		SMS, Email, or Phone. Determines which channel reminders use.
is_active	BOOLEAN		Soft delete. Inactive clients excluded from search but historical data preserved.

vehicles

Vehicle records linked to clients via a junction table. A vehicle can have multiple owners over time, and an owner can have multiple vehicles.

Field	Type	Tags	Validation and notes
vehicle_id	VARCHAR(10)	PK, Auto	Format: VH-001.
registration	VARCHAR(20)	Required, Unique	NSW format. Primary lookup field — mechanics search by rego.
make_id	VARCHAR(10)	FK	Links to vehicle_makes. 77 makes pre-loaded.
model/year/colour	Various		Auto-populated via rego lookup API in phase 2. Currently 99% empty in Dhalla data.
vin	VARCHAR(30)	Validated	17-character VIN. Globally unique vehicle identifier.

Field	Type	Tags	Validation and notes
current_mileage_km	INTEGER		Updated each time the vehicle visits. Used for service interval calculations.
next_service_due	DATE	Auto	Auto-set from invoice: last_service_date + interval (default 6 months).
next_service_km	INTEGER	Auto	Auto-set from invoice: current_mileage + interval (default 10,000 km).
pink_slip_expiry	DATE		NSW e-safety check expiry. Drives pink slip reminder cadence.

client_vehicles (junction)

Many-to-many link between clients and vehicles, with ownership dates and relationship types.

staff

Workshop employees. Tracks certifications (MVRL for pink slips, ARC for AC work), hourly rates, and availability.

users

Login accounts with role-based access: Owner, Manager, Front Desk, Mechanic, Read-Only. Each role has a defined permission matrix controlling access to financial data, settings, and sensitive operations.

4.2 Service catalogue

job_categories

12 top-level groupings: Engine Service, Brake System, Transmission, Cooling, Electrical, Suspension & Steering, Tyres & Wheels, Exhaust, HVAC/AC, Inspections, General Labour, Body & Exterior.

jobs (parent jobs)

25 parent jobs within the 12 categories. Each has a standard_total_price (sum of sub-items) and estimated_duration. Examples: Minor Service (\$154, 1.0hr), Front Brake Replacement (\$280, 1.5hrs), Pink Slip (\$52, 0.5hr).

job_sub_items

40+ atomic work items. Each answers three questions: what part does it use, how long does it take, and what does it cost? Calculation chain: labour_time_hrs × labour_rate_hr = labour_cost + part_cost = standard_price. Sum of sub-item standard_prices = parent job total.

job_item_parts (linking table)

Links sub-items to inventory parts with default quantities. When a parent job is selected on a quote, the system pre-populates all sub-items and their linked parts. On job completion, linked parts auto-deduct from stock.

4.3 Workflow pipeline

quotations + quote_lines

Entry point of revenue pipeline. 6 statuses: Draft → Sent → Approved/Declined/Expired → Converted. SMS approval: customer replies YES to approve. Quote validity: configurable (default 14 days). Supplementary quotes for discovered work.

job_cards + job_card_lines

Active work tracker. 8 statuses: Booked → Waiting → In Progress → Waiting for Parts → QC → Ready for Pickup → Completed → Cancelled. Assigned to mechanic and bay. Progress tracked per sub-item with actual_labour_hrs.

invoices + invoice_lines

ATO-compliant tax invoices. Sequential numbering (no gaps). Per-line GST calculation. Auto-generated from completed job cards. PDF with workshop branding, ABN, vehicle details, line items, and GST breakdown.

payments

Tracks every dollar in and out. Types: Customer, Supplier, Refund, Expense. Methods: Cash, Credit Card, Debit Card, EFTPOS, PayID, Bank Transfer, BPay. Supports partial payments and multi-method on single invoice.

4.4 Operations

parts + part_categories + suppliers + purchase_orders

123 parts across 11 categories. Cost and sell prices with margin visibility. 5 stock movement types (PO received, return to stock, job completed, wastage, stocktake adjustment). Reorder engine with 6 alert levels. Auto-draft PO generation. 5 suppliers with credit terms and lead times.

appointments

Booking system with bay assignment, mechanic allocation, capacity calculation, confirmation tracking. Sources: Phone, Walk-in, SMS Reply, Customer Portal, Website.

service_reminders

276 active reminders. Types: Next Service, Pink Slip. Multi-touch cadence: 30/14/7 days before due. Pink slips start at 60 days. Two-way SMS with keyword parsing: YES/BOOK creates appointment, STOP opts out.

maintenance_history

Immutable service records auto-generated from completed job cards. Vehicle-bound (stays with vehicle through ownership changes). Includes service type, mileage, cost, mechanic, notes.

accounting_ledger

Unified transaction record. Auto-populated from invoices, POs, and payments. Drives BAS/GST calculations and all financial reporting. Append-only for audit integrity.

audit_log

Every create, update, delete action logged. Includes user, timestamp, table, record, old/new values, IP address. Immutable. 7-year ATO retention.

5. Module specifications

Each module is fully specified in the interactive deep-dive widgets delivered during the design phase. Below is a summary of the business rules, edge cases, and critical logic for each module. The interactive specifications remain the authoritative reference for field-level detail.

5.1 Module 1: Core entities

Workshops, clients (Individual/Business with progressive data capture), vehicles (rego lookup API auto-fill), client-vehicle junction (one client to many vehicles), staff (roles, hourly rates, MVRL/ARC certifications), users (5 roles with full permission matrix: Owner, Manager, Front Desk, Mechanic, Read-Only).

5.2 Module 2: Service catalogue

3-tier hierarchy: 12 categories → 25 parent jobs → 40+ sub-items. 5-layer pricing engine (workshop rate → sub-item → package → quote override → invoice adjustment). GST rules (ex-GST storage, per-line calculation at 10%). Job-to-parts linking table enables auto-populated quotes and inventory auto-deduction. Discount logic with maximum without owner approval (default 15%). Margin visibility for Owner/Manager roles.

5.3 Module 3: Parts and inventory

123 parts across 11 categories. Dual pricing (cost and sell) with margin visibility. 5 stock movement types: PO received, return to stock, job completed, wastage, stocktake adjustment. Reorder engine with 6 alert levels (healthy, getting low, reorder now, critical, out of stock, on demand). Auto-draft PO generation. Supplier management with credit terms and lead times. Location system (Shelf A1-F2, Tyre rack). Barcode scanning workflow (Code 128, phase 2).

5.4 Module 4: Workflow pipeline

Quote → Job Card → Invoice → Payment lifecycle. Quote: 6 statuses with SMS approval ("Reply YES to approve"). Job card: 8-status lifecycle with bay assignment and per-item progress tracking. Invoice: ATO-compliant with sequential numbering, per-line GST, branded PDF generation. Payment: 6 methods, partial payment support, multi-method per invoice. Shared line item structure across all stages (copy, not move). Supplementary quote workflow for discovered extra work.

5.5 Module 5: Scheduling and reminders

Appointments with capacity calculation (bays × hours). Bay management floor board (wall-mounted screen view with live status, progress bars, mechanic assignment). Service reminder cadence: 30/14/7 days before due, overdue at +30 days. Pink slip tracking: 60/30/14 days before expiry, MVRL certification enforcement. Two-way SMS with keyword parsing (YES, BOOK, STOP, C, R). ACMA compliance (9am-8pm send window, opt-out, frequency cap). Daily morning briefing view.

5.6 Module 6: Accounting and reporting

Unified transaction ledger auto-populated from invoices, POs, payments. BAS/GST quarterly calculation for Australian FY (Jul-Jun, Q1-Q4). Cash vs accrual accounting support. 10 structured reports: revenue summary, P&L, aged receivables, accounts payable, mechanic productivity, parts profitability, customer value, inventory valuation, BAS summary, daily takings. Dashboard KPIs computed live. Export roadmap: Phase 1 CSV, Phase 2 Xero API, Phase 3 MYOB/QuickBooks.

5.7 Module 7: Maintenance history and vehicle lifecycle

Immutable service records auto-generated from completed job cards. Vehicle timeline with colour-coded events (service, inspection, significant repair, ownership change, mileage milestone). Customer portal with magic-link access (no login required), service history, invoice PDF downloads, booking CTA, recommendation banners. Progressive data completeness tracking (clients 27%, vehicles 6%, parts 15% at launch). Audit trail (append-only, 7-year ATO retention). Vehicle ownership transfer workflow.

5.8 Module 8: SaaS platform

Multi-tenancy with shared DB and PostgreSQL RLS. Onboarding wizard (7 steps, first invoice in 20 minutes). Workshop settings (10 categories: business, invoicing, pricing, scheduling, SMS, reminders, users, billing, integrations, branding). 3-tier subscription (\$99/\$199/\$349 per month). Data migration wizard (CSV upload with column mapping, duplicate detection, validation). White-label branding (custom domain, custom login, branded PDFs, custom SMS sender). 14-day free trial with full Professional features.

6. UI screen prototypes

12 interactive React/JSX prototypes have been delivered across 4 files. These are not wireframes — they are functional React components with real data, working interactions, and the production design system (Navy #1B2A4A + Amber #E8920D palette, JetBrains Mono for regos/invoice numbers, DM Sans for body text, 12px radius cards). Each prototype demonstrates the exact layout, data flow, and interaction patterns for production implementation.

6.1 Design system

Token	Value	Usage
Primary navy	#1B2A4A	Sidebar, headers, invoice header, primary text emphasis
Accent amber	#E8920D	CTAs, active states, highlights, links, amber badges
Background	#F3F5F7	Page background
Card white	#FFFFFF	Card backgrounds, modal backgrounds
Border	#E5E7EB	Card borders, table dividers, input borders
Success green	#059669	Paid status, healthy stock, positive trends
Danger red	#DC2626	Overdue, out of stock, critical alerts
Warning amber	#D97706	Unconfirmed, getting low, in progress
Info blue	#2563EB	Booked, informational badges
Monospace font	JetBrains Mono	Registration numbers, invoice IDs, part numbers, prices
Body font	DM Sans / Inter	All body text, labels, buttons
Card radius	12px	All card containers
Control radius	8px	Buttons, inputs, badges

6.2 Screen inventory

Screen	File	Description
Dashboard	MechWise_Dashboard_UI.jsx	Sidebar nav, KPIs, floor board, schedule, alerts, revenue tracker
Client detail	MechWise_Screens_UI.jsx	Contact card, vehicle list, stats, service history timeline
Quote builder	MechWise_Screens_UI.jsx	Catalogue picker, line items with sub-item table, GST calculator, SMS/email delivery
Job card	MechWise_Screens_UI.jsx	Progress checklist, time tracking, status lifecycle selector, supplementary quote
Invoice view	MechWise_Screens_2_UI.jsx	PDF-style layout with navy header, ATO-compliant format, line items, GST
Payment modal	MechWise_Screens_2_UI.jsx	6 payment methods (EFTPOS, Cash, Credit Card, PayID, Bank Transfer, BPay)
Inventory	MechWise_Screens_2_UI.jsx	Category sidebar, stock table with margin %, status badges, alerts
Calendar	MechWise_Screens_2_UI.jsx	Day view with bay columns, appointment blocks, utilisation bar
Reminders	MechWise_Screens_3_UI.jsx	Filter tabs, bulk SMS actions, KPI tiles, send tracking
Reports	MechWise_Screens_3_UI.jsx	Revenue bar chart, job-type breakdown, mechanic productivity, top customers
Settings	MechWise_Screens_3_UI.jsx	Business details form, SMS config, template editor, logo upload

Screen	File	Description
Customer portal	MechWise_Screens_3_UI.jsx	Vehicle selector, status cards, booking CTA, service history, invoice downloads

7. Development plan

7.1 Phase 1 — MVP (8 weeks)

Goal: Replace Dhalla's current PHP system. Every feature the old system had, plus critical improvements.

Deliverable	Effort	Dependencies
Auth + multi-tenancy + DB schema deployment	1 week	None — start here
Client and vehicle CRUD + search	1.5 weeks	Auth
Service catalogue (categories, jobs, sub-items, pricing)	1 week	DB schema
Quote builder → job card → invoice pipeline	2 weeks	Clients, vehicles, catalogue
Payment recording + history	0.5 week	Invoices
Parts inventory + stock management + alerts	1 week	DB schema
Data migration tool (CSV import for Dhalla)	0.5 week	Clients, vehicles, parts
Invoice PDF generation (ATO-compliant)	0.5 week	Invoices

7.2 Phase 2 — Growth (6 weeks)

Goal: Automate customer retention and workshop operations.

Deliverable	Effort	Dependencies
SMS automation engine + templates + two-way SMS	2 weeks	Clients, reminders, SMS provider
Appointment calendar + bay columns + booking form	1.5 weeks	Bays, staff
Workshop floor board (live bay status)	1 week	Job cards, bays
Dashboard with KPIs + alerts + schedule	1 week	All pipeline data
Supplier management + PO creation + receipt	0.5 week	Parts, suppliers

7.3 Phase 3 — Scale (6 weeks)

Goal: Reports, portal, and billing. Product ready for general sale.

Deliverable	Effort	Dependencies
Reports (revenue, P&L, productivity, BAS)	2 weeks	Accounting ledger
Customer portal (magic-link, history, downloads)	1.5 weeks	Clients, invoices, vehicles
Settings panel (all workshop configuration)	1 week	Workshop table
Accounting export (CSV for Xero/MYOB, BAS PDF)	0.5 week	Ledger, BAS calculations
Onboarding wizard for new workshops	0.5 week	Workshop, catalogue defaults
Subscription billing (Stripe integration)	0.5 week	Stripe account

7.4 Phase 4 — Enterprise (ongoing)

Multi-workshop support, Xero API two-way sync, rego lookup API, white-label branding, mobile PWA optimisation.

8. Australian compliance requirements

8.1 ATO invoicing requirements

- Invoice must include supplier ABN (95611566888 for Dhalla).
- Sequential invoice numbering with no gaps. Voided invoices stay in sequence.
- Per-line GST calculation at 10%. Prices stored ex-GST internally.
- Must clearly state "Tax Invoice" on the document.
- Vehicle details (rego, make, model, odometer) on the invoice.
- 7-year record retention for all financial documents.

8.2 BAS/GST compliance

- Australian financial year: July to June.
- BAS quarters: Q1 Jul-Sep, Q2 Oct-Dec, Q3 Jan-Mar, Q4 Apr-Jun.
- GST on sales (1A) calculated from invoice line items.
- GST on purchases (1B) calculated from PO line items and expenses.
- Net GST payable (9) = 1A - 1B.
- Cash basis (default): report GST when money changes hands.
- Accrual basis (optional): report GST when invoice is issued.

8.3 NSW pink slip (e-safety) rules

- Required annually for vehicles over 5 years old in NSW.
- Workshop must hold MVRL to conduct inspections.
- E-safety certificate issued via DRIVES system (NSW government).
- Certificate valid for 6 months for rego renewal purposes.
- System enforces: only MVRL-certified staff can complete pink slip job cards.

8.4 ACMA SMS compliance (Spam Act 2003)

- Send window: 9:00am to 8:00pm local time (AEST/AEDT).
- Opt-out mechanism required: "Reply STOP to opt out" in every message.
- Frequency cap: max 1 SMS per customer per week (excluding direct replies).
- Service reminders are transactional (existing customer relationship) — exempt from some provisions but opt-out still required.

9. Dhalla Automotive — reference data

The first client's data has been cleaned, deduplicated, and structured for migration. This data populates the demo environment and serves as the test dataset during development.

Entity	Record count	Notes
Clients	300 (deduplicated from 323)	Matched by phone number. 88 missing phone numbers. 0 emails captured.
Vehicles	323	Registration numbers captured. Make/model/year mostly empty (99%). Rego lookup will fix.
Client-vehicle links	323	One-to-one initially. Multi-vehicle owners identified for manual review.
Vehicle makes	77	Pre-loaded brand list.
Staff/mechanics	11	Names captured. Hourly rates, certifications, phones need entry.
Job categories	12	Restructured from flat list of 58 items.
Parent jobs	25	Restructured from flat list. Prices and durations set.
Job sub-items	40+	Decomposed from parent jobs with part linkage.
Part categories	11	Organised by physical type.
Parts	123	Sell prices captured. Cost prices, stock levels, suppliers mostly missing.
Service reminders	212	Next service alerts with due dates. Many overdue.
Pink slip reminders	64	With expiry dates.

10. Cost estimates and revenue projections

10.1 Build cost options

Option	Team	Cost	Timeline
A: Solo dev + AI	1 senior contractor + you (product owner) + Claude	\$78k-116k	24 weeks
B: Small team	1 senior lead + 1 junior frontend + part-time designer + you	\$137k-189k	20 weeks
C: Build with Claude Code	You (full-time) + Claude Code + spot designer hire	\$5k-7k	24 weeks

10.2 Ongoing costs (monthly)

Service	Cost	Notes
Vercel Pro	\$20/month	Frontend hosting with preview deployments
Railway	\$20-40/month	API + PostgreSQL + Redis
SMS (MessageMedia)	\$35/month	~500 messages at \$0.07 each
Postmark email	\$15/month	Transactional email
Sentry	\$0 (free tier)	Error tracking — free for <5k events/month
Domain + DNS	\$15/year	mechwise.com.au
Total at launch	~\$100/month	Scales with usage

10.3 Revenue projections

Metric	Year 1	Year 2
Target workshops	10	30
Average subscription	\$199/month	\$220/month
Annual recurring revenue	\$24k-48k	\$72k-144k
Setup fees	\$5k-15k	\$15k-45k
Break-even (Option A)	15-20 workshops	Already profitable

11. Deliverable file index

File	Type	Description
MechWise_Enhanced_Database_v2.xlsx	Excel	26-sheet database workbook with all real Dhalla data populated
MechWise_Database_Schema.sql	SQL	Full PostgreSQL schema: 20 tables, 30+ indexes, 6 views, FK constraints, enums
MechWise_Dashboard_UI.jsx	React	Dashboard with sidebar nav, KPIs, floor board, schedule, alerts, revenue tracker
MechWise_Screens_UI.jsx	React	Client detail, quote builder, job card screens
MechWise_Screens_2_UI.jsx	React	Invoice, payment modal, inventory management, calendar/scheduling
MechWise_Screens_3_UI.jsx	React	Service reminders, reports & analytics, settings, customer portal
MechWise_Technical_Spec.docx	Word	This document — consolidated technical specification

End of document.